Attention-Driven Keyword Extraction for Conversational Retrieval-Augmented Generation

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Abstract

Conversational retrieval systems commonly employ Large Language Models (LLMs) to reformulate historical dialogues for query understanding. However, this approach frequently suffers from critical keyword omission and hallucination issues, leading to retrieval failures. In this paper, we propose a novel framework for conversational Retrieval-Augmented Generation (RAG) that leverages the attention mechanism during LLM's reformulation token generation to identify and extract salient keywords from dialogue history. Our method performs denoising on the historical conversations for more effective retrieval while simultaneously compressing the conversational context appropriately to accommodate long-context input during the generation phase. A key advantage of our approach is that it introduces no additional computational overhead while achieving substantial performance improvements. By exploiting the intrinsic attention patterns of LLMs, our framework effectively bridges the gap between query reformulation and retrieval accuracy. Furthermore, we conduct an in-depth interpretability analysis of LLMs, examining how different layers and attention heads contribute to keyword selection and information retrieval tasks. This analysis provides valuable insights into the internal mechanisms of LLMs in conversational search scenarios. Experimental results demonstrate that our attention-based keyword extraction mechanism significantly outperforms traditional reformulation methods, mitigating both information loss and hallucination problems. The proposed context compression strategy further enhances the overall system efficiency without compromising generation quality, making our approach a practical and effective solution for conversational search systems.

CCS CONCEPTS

Information systems ~ Information retrieval ~ Information retrieval query processing

Keywords

Conversational Retrieval, Query Reformulation, Attention Mechanism, Retrieval-Augmented Generation (RAG)

1 INTRODUCTION

Conversational search systems have emerged as a transformative paradigm in information retrieval, enabling users to interact with search engines through natural multi-turn dialogues rather than isolated keyword queries[1]. Unlike traditional search engines, conversational systems allow users to engage in contextual exchanges, progressively refining their information needs through iterative interactions. However, a fundamental challenge in conversational search lies in accurately capturing user intent across dialogue turns, where queries often contain linguistic phenomena such as coreference, ellipsis, and context-dependent expressions [2].

To address the contextual understanding challenge, existing approaches typically employ Large Language Models (LLMs) to reformulate conversational queries into self-contained, context-independent forms suitable for retrieval. This reformulation process, while intuitive, frequently encounters two critical limitations: (1) **keyword omission**, where essential terms from the dialogue history are inadvertently excluded during reformulation, and (2) **hallucination**, where the model generates plausible but factually incorrect or irrelevant content[3]. These issues can severely degrade retrieval effectiveness, as the reformulated queries fail to accurately represent the user's true information need.

Recent advances in Retrieval-Augmented Generation (RAG) have demonstrated the potential of combining retrieval and generation capabilities for conversational AI systems[4]. However, the integration of conversational context into RAG pipelines remains challenging, particularly when dealing with long dialogue histories that exceed the context window limitations of generation models. Moreover, existing methods often treat reformulation and retrieval as separate, independent stages, missing opportunities to leverage the intrinsic knowledge encoded within LLMs' internal representations.

In this work, we propose a novel framework that exploits the attention mechanism of LLMs during query reformulation to enhance conversational retrieval in RAG systems. Our key insight is that the attention patterns generated when LLMs produce reformulation tokens inherently encode information about which historical dialogue elements are most relevant for retrieval. By analyzing these attention distributions, we can systematically identify and extract salient keywords from the dialogue history while filtering out noisy or irrelevant information. Specifically, our method performs denoising on historical conversations to construct more precise retrieval queries, while simultaneously compressing the conversational context to accommodate the long-context requirements of the generation phase.

A distinctive advantage of our approach is its **zero additional computational cost**—we leverage attention information that is already computed during the standard LLM forward pass, requiring no extra model parameters or inference overhead. Furthermore, we conduct an in-depth **interpretability analysis** of LLMs in the context of conversational retrieval, systematically examining how different transformer layers and attention heads contribute to keyword selection and information retrieval tasks. This analysis provides valuable insights into the internal mechanisms by which LLMs process conversational context and reveals which model components are most effective for retrieval-oriented tasks.

Our framework addresses several critical gaps in existing conversational search systems:

First, unlike traditional query reformulation methods that treat LLM outputs as black boxes, we explicitly leverage the attention mechanism to guide keyword extraction, providing a more interpretable and controllable approach to query understanding.

Second, by performing denoising on dialogue history rather than solely relying on generated reformulations, we mitigate both keyword omission and hallucination issues that plague existing methods.

Third, our context compression strategy enables efficient handling of long conversational histories while preserving essential information for both retrieval and generation stages.

Fourth, through systematic interpretability analysis across different layers and attention heads, we provide actionable insights for designing more effective conversational retrieval architectures.

We conduct extensive experiments on multiple conversational search benchmarks to validate the effectiveness of our approach. The results demonstrate that our attention-based framework significantly outperforms traditional reformulation methods, achieving substantial improvements in retrieval accuracy while maintaining generation

quality. The interpretability analysis further reveals interesting patterns in how LLMs allocate attention to different dialogue elements, offering new perspectives on model behavior in conversational scenarios.

In summary, our main contributions are:

• **A novel attention-based framework** for conversational RAG that leverages LLM attention mechanisms to extract salient keywords and denoise dialogue history, achieving superior retrieval performance without additional computational cost.

• **An effective context compression strategy** that balances the trade-off between preserving conversational context and accommodating long-context input requirements for generation models.

• **Comprehensive interpretability analysis** examining how different LLM layers and attention heads contribute to keyword selection and information retrieval, providing insights into the internal mechanisms of conversational understanding.

• **Extensive experimental validation** demonstrating significant performance improvements over state-of-the-art baselines on multiple conversational search benchmarks, with detailed ablation studies confirming the effectiveness of each component.

2 RELATED WORK

Transformer and Attention Mechanism

The Transformer architecture[5] has revolutionized natural language processing by introducing the self-attention mechanism as a fundamental building block for sequence modeling. Unlike recurrent neural networks that process sequences sequentially, Transformers enable parallel computation through multi-head self-attention, which allows each token to attend to all other tokens in the sequence simultaneously. This mechanism computes attention weights that represent the relevance between different positions, enabling the model to capture long-range dependencies effectively.

The attention mechanism in Transformers has been extensively studied for its interpretability and utility beyond standard sequence modeling tasks. Clark et al.[6] demonstrated that different attention heads in BERT learn to capture distinct linguistic phenomena, such as syntactic dependencies and coreference relationships. Voita et al. [7] further showed that many attention heads can be pruned without significant performance degradation, suggesting that different heads play varying roles in model behavior.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm that combines the strengths of neural retrieval systems and large language models for knowledge-intensive tasks[8]. RAG systems first retrieve relevant documents from a knowledge base and then condition the generation process on these retrieved contents, enabling models to access up-to-date information beyond their training data and reducing hallucination issues.

Several variants of RAG have been proposed to improve different aspects of the retrieval-generation pipeline. Izacard and Grave[9] introduced Fusion-in-Decoder (FiD), which independently encodes multiple retrieved passages and fuses them during decoding. Shi et al.[10] proposed REPLUG, which treats the language model as a black box and uses retrieved documents to refine output distributions. Asai et al.[11] developed Self-RAG, which enables models to self-reflect on retrieved passages and their own generations through special reflection tokens.

Despite these advances, most RAG systems are designed for single-turn question answering scenarios and do not adequately address the challenges of conversational contexts. Recent work has begun exploring conversational RAG[12,13], but these methods typically rely on simple query reformulation or context concatenation strategies that

may introduce noise or lose critical information. Moreover, existing approaches often treat retrieval and generation as loosely coupled components, missing opportunities to leverage the intrinsic knowledge encoded in LLM representations. Our work addresses these limitations by proposing an attention-based framework that tightly integrates conversational understanding with both retrieval and generation stages.

Query Expansion and Query Rewrite in Conversational Search

Conversational search aims to support information seeking through multi-turn natural language dialogues, where users progressively refine their information needs through iterative interactions[14]. A fundamental challenge in conversational search is understanding context-dependent queries that contain linguistic phenomena such as coreference, ellipsis, and topic shifts[15-17].

Query expansion, a classical technique in information retrieval, aims to enhance effectiveness by augmenting the original query with contextually relevant terms[18]. In conversational settings, various strategies have been employed to adapt this technique. Relevance term classification approaches recast query reformulation as a binary classification task to identify and append salient terms from the dialogue context. For instance, Kumar et al. [19] utilized weakly-supervised signals to train classifiers for term selection, combining this with pseudo-relevance feedback (PRF) for refinement. Voskarides et al. [20] introduced QuReTeC, which leverages distant supervision and pseudo-labels inferred from user-system interactions to guide term selection. Similarly, Lin et al. [21,22] developed term importance estimation models, which rank terms within context by their significance and ensure retention of critical information during reformulation. Beyond term-level selection, some approaches focus on turn-level query selection to identify which historical queries are most relevant for expanding the current query. Notably, Mo et al. [23] proposed a learning-based method to relate current queries to previous conversational turns, addressing the challenge that not all historical queries are equally useful for query expansion.

Conversational Query Rewriting (CQR) approaches address this challenge by transforming context-dependent queries into self-contained, de-contextualized forms suitable for standard retrieval systems. Early methods employed sequence-to-sequence models trained on human-rewritten queries[24]. The advent of large language models has opened new possibilities for conversational query understanding. Mo et al. [25] proposed ConvGQR, which uses LLMs to generate multiple query reformulations and hypothetical responses to enrich search intent representation. Mao et al. [26] introduced LLM4CS, a prompting framework that leverages LLMs as search intent interpreters by generating query rewrites and hypothetical responses, then fusing their representations to enhance conversational search performance. Wu et al. [27] demonstrated that human-annotated rewrites are not necessarily optimal for search. To address this, several studies[28] have optimized rewriting models directly for downstream search tasks by utilizing retrieval performance as a reward signal. Furthermore, Ye et al. designed comprehensive multi-stage rewriting instructions and employed a rewrite-then-edit strategy to generate high-quality query rewrites. Despite these advances, LLM-based reformulation methods still suffer from keyword omission and hallucination issues, as the generated rewrites may inadvertently exclude critical terms or introduce irrelevant content. However, these methods typically treat different components as independent modules and do not exploit the internal representations of LLMs for retrieval optimization.

Positioning of Our Work

Our work bridges several research areas by proposing a novel framework that exploits LLM attention mechanisms for conversational RAG. Unlike existing query reformulation methods that treat LLM outputs as black boxes, we explicitly leverage attention patterns to identify salient keywords and denoise dialogue history. Compared to traditional RAG systems designed for single-turn scenarios, our approach addresses the unique

challenges of multi-turn conversations through attention-based context compression and keyword extraction. Furthermore, our interpretability analysis provides new insights into how LLMs process conversational contexts, contributing to the broader understanding of these models in information retrieval applications. Importantly, our framework achieves these improvements without introducing additional computational overhead, making it practical for real-world deployment.

3 METHODOLOGY

3.1 OVERVIEW

Figure 1 illustrates the overall architecture of our proposed attention-based conversational retrieval framework. Given a multi-turn dialogue history $H = \{(q_1, r_1), (q_2, r_2), \dots, (q_{n-1}, r_{n-1})\}$ and the current query q_n, our system performs three key operations: (1) Query Reformulation using LLMs to generate a self-contained query q'_n (2) Attention-based Keyword Extraction from dialogue history by analyzing the attention patterns during reformulation token generation, and (3) Hybrid Retrieval that combines reformulated queries with extracted keywords for enhanced retrieval accuracy. Crucially, our method exploits the already-computed attention mechanisms in LLMs, introducing zero additional computational overhead while significantly improving retrieval effectiveness.

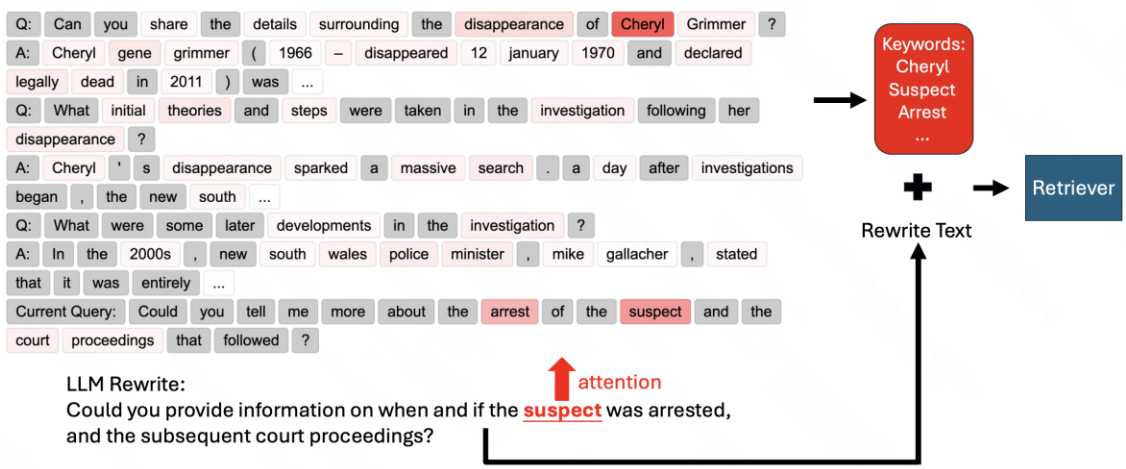

Figure 1. Illustration of Attention-driven Keyword Selection from Dialogue History. Gray-highlighted words represent stopwords excluded from retrieval. Color intensity (white to red) indicates attention scores from the output token "suspect" to historical words, with higher scores shown in darker red. Extracted high-attention keywords (e.g., "Cheryl", "suspect", "arrest") are concatenated with the rewritten query for retrieval. Notably, "suspect" refers to the suspect in Cheryl's case rather than Cheryl herself, demonstrating that the attention mechanism captures complex semantic relationships beyond simple coreference.

3.2 Query Reformulation with LLM

Following standard practice in conversational search, we employ an LLM to reformulate the current query q_n into a context-independent form q'_n by conditioning on the dialogue history. Formally, the reformulation process can be expressed as:

$$q'_n = \mathrm{LLM}(q_n | H) \qquad (1)$$

where the LLM generates a sequence of tokens $q'_n = \{t_1, t_2, \dots, t_m\}$ through autoregressive decoding. During this generation process, the transformer-based LLM computes multi-head attention across all input tokens (dialogue history and current query) at each layer. Let $\boldsymbol{A}_{ij}^{(l,h)}$ denote the attention weight from the i-th output token to the j-th input token at layer l and attention head h:

$$A_{ij}^{(l,h)} = \text{softmax}\left(\frac{Q_i^{(l,h)} \cdot \left(K_j^{(l,h)}\right)^T}{\sqrt{d_k}}\right) \qquad (2)$$

where $Q_i^{(l,h)}$ and $K_j^{(l,h)}$ represent the query and key vectors at layer l and head h, and d_k is the dimension of the key vectors.

$$H_{\text{compressed}} = \{(q_i, r_i) \mid \textstyle\sum_{j \in (q_i, r_i)} s_j > \theta\} \qquad (3)$$

3.3 Attention-Based Keyword Extraction

3.3.1Attention Aggregation Strategies.

The core innovation of our approach lies in leveraging these attention patterns to identify salient keywords from the dialogue history. We investigate three aggregation strategies to compute keyword importance scores:

Layer-Averaged Attention. We first compute the average attention score across all attention heads within a specific layer l:

$$\mathrm{A}_{ij}^{(l)} = \frac{1}{H} \textstyle\sum_{h=1}^{H} \mathrm{A}_j^{(l,h)} \qquad (4)$$

where H is the number of attention heads. For each input token j from the dialogue history, we aggregate attention scores from all output tokens:

$$s_j^{(l)} = \textstyle\sum_{i=1}^{m} \mathrm{A}_j^{(l)} \qquad (5)$$

This yields a keyword importance score for each historical token at layer l.

Expert Head Selection. Inspired by the observation that different attention heads specialize in different linguistic phenomena, we identify "expert heads" that are particularly effective for keyword extraction. For each attention head (l, h), we compute:

$$\mathrm{s}_{\mathrm{j}}^{(\mathrm{l,h})} = \textstyle\sum_{\mathrm{i}=1}^{\mathrm{m}} \mathbf{A}_{\mathrm{ij}}^{(\mathrm{l,h})} \qquad (6)$$

We then evaluate each head's performance on a validation set and select the top-K heads with highest retrieval accuracy as expert heads $\mathcal{E} = \{(l_1, h_1), (l_2, h_2), \dots, (l_K, h_K)\}$.

Expert Ensemble. The final keyword importance score combines evidence from multiple expert heads:

$$s_j = \frac{1}{|\mathcal{E}|} \textstyle\sum_{(l,h) \in \mathcal{E}} s_j^{(l,h)} \qquad (7)$$

3.3.2 Top-K Keyword Selection.

Instead of using global importance scores, we extract keywords from each selected output token individually. For each output token $i \in \mathcal{T}_{\text{selected}}$, we select the top-K dialogue history tokens with highest attention weights:

$$\mathcal{K}_i = \mathrm{top}_{\mathrm{K}}\left(\{w_j\}, A_{ij}\right) \qquad (8)$$

where w_j denotes the word corresponding to token j. The final keyword set is obtained by aggregating and deduplicating keywords across all selected output tokens:

$$\mathcal{K} = \textstyle\bigcup_{i \in \mathcal{T}_{\text{selected}}} K_i \qquad (9)$$

This approach allows each output token to contribute its most relevant keywords, while deduplication ensures efficiency. We investigate different values of K to balance between recall (capturing sufficient context) and precision (avoiding noise).

3.3.3 Output Token Selection via Attention Concentration.

A critical question is: which output tokens should be used for keyword extraction? We propose a novel criterion based on attention concentration, measured by the Attention Entropy (AE) of attention distributions. Interestingly, our experimental findings reveal that output tokens with diffused attention patterns (high entropy) are more effective for identifying relevant keywords, while tokens with concentrated attention (low entropy) tend to be overly specific and miss important contextual information necessary for retrieval tasks.

For each output token i, we compute the attention entropy across dialogue history tokens, excluding stopwords:

$$\widetilde{\boldsymbol{A}_{ij}^{(l,h)}} = \frac{\boldsymbol{A}_{ij}^{(l,h)}}{\sum_{k \in \mathcal{D}_{\text{non-stop}}} \boldsymbol{A}_{ik}^{(l,h)}}, \quad j \in \mathcal{D}_{\text{non-stop}} \qquad (10)$$

where $\mathcal{D}_{\text{non-stop}}$ denotes the set of non-stopword tokens in the dialogue history. We then compute the attention entropy using the normalized attention distribution:

$$\mathcal{H}_i^{(l,h)} = -\sum_{j \in \mathcal{D}_{\text{non-stop}}} \widetilde{\boldsymbol{A}_{ij}^{(l,h)}} \log \widetilde{\boldsymbol{A}_{ij}^{(l,h)}} \qquad (11)$$

This normalization ensures that attention distributions are on a comparable scale across different tokens, making the entropy values directly comparable. The AE metric reflects how concentrated a token's attention is over the preceding conversational context. Higher entropy indicates that the token attends to a broader range of dialogue content, capturing more comprehensive contextual information.

To adapt this mechanism for retrieval tasks, we select the top-α percent of output tokens with highest entropy:

$$\mathcal{T}_{\text{selected}} = \text{top}_{\alpha}\left(\{t_i\}, \mathcal{H}_{ij}\right) \qquad (12)$$

Keyword extraction then uses only the attention patterns from these selected output tokens:

$$s_j = \sum_{i \in \mathcal{T}_{\text{selected}}} \mathbf{A}_{ij} \qquad (13)$$

This concentration-based selection strategy effectively filters out uninformative tokens that may hallucinate or attend indiscriminately to the entire history.

3.4 Hybrid Retrieval Strategy

The extracted keywords $\mathcal{K}$ and the reformulated query q_n' provide complementary information for retrieval. We explore two hybrid strategies:

Direct Keyword Retrieval. Use extracted keywords directly as retrieval queries:

$$\mathcal{D}_{\text{keywords}} = \text{Retrieve}(\mathcal{K}) \qquad (14)$$

Concatenated Retrieval. Combine reformulated query with extracted keywords:

$$q_n'' = q_n' \oplus \mathcal{K} \quad (15)$$

$$\mathcal{D}_{\text{hybrid}} = \text{Retrieve}(q_n'') \quad (16)$$

where $\oplus$ denotes concatenation. This hybrid approach mitigates keyword omission in reformulation while reducing hallucination through grounding in actual dialogue tokens.

3.5 Context Compression for Generation

To accommodate long dialogue histories within the context window constraints of generation models, we leverage the same attention-based importance scores for context compression. Specifically, we retain dialogue turns and segments with high cumulative keyword importance:

$$H_{\text{compressed}} = \{(q_i, r_i) \mid \sum_{j \in (q_i, r_i)} s_j > \theta\} \qquad (17)$$

where θ is a threshold tuned on validation data. This ensures that the most retrieval-relevant context is preserved for the generation phase, while redundant or less informative dialogue segments are pruned.

3.6 Interpretability Analysis

To gain deeper insights into how LLMs process conversational context, we conduct systematic interpretability analyses across three dimensions:

Layer-wise Analysis. We evaluate the keyword extraction performance of each layer independently to understand how information flows through the transformer stack. This reveals which layers encode the most useful representations for retrieval tasks.

Head-wise Analysis. We analyze individual attention heads to identify specialized heads that consistently attend to retrieval-relevant tokens. By comparing head behavior across different model scales, we investigate whether certain attention patterns emerge universally or are model-specific.

Token-wise Analysis. We examine the relationship between output token characteristics (entropy, position, semantic content) and their effectiveness for keyword extraction. This provides guidance on which tokens should be prioritized during attention aggregation.

3.7 Computational Efficiency

A key advantage of our framework is that it introduces zero additional computational cost during inference. The attention weights $A_{ij}^{(l,h)}$ are already computed during the standard LLM forward pass for query reformulation. Our method simply leverages these existing attention patterns for keyword extraction, requiring only lightweight post-processing operations (summation, sorting, thresholding) that are negligible compared to LLM inference time. This makes our approach highly practical for real-world conversational search systems where computational efficiency is critical.

In summary, our methodology provides a principled framework for exploiting LLM attention mechanisms to enhance conversational retrieval. By systematically analyzing attention patterns across layers, heads, and tokens, we develop an effective strategy for keyword extraction that mitigates both omission and hallucination issues while maintaining computational efficiency.

4 EXPERIMENTS

4.1 Experimental Setup

4.1.1Dataset and Evaluation Metrics.

We conduct experiments on the CORAL[12] dataset, a comprehensive benchmark for conversational search and retrieval-augmented generation. CORAL contains multi-turn dialogues with information-seeking queries requiring contextual understanding across conversation history.

Retrieval Metrics: We employ standard retrieval evaluation metrics:

MRR: Mean Reciprocal Rank, evaluating the rank of the first relevant document

MAP: Mean Average Precision, measuring the average precision across all relevant documents

NDCG@3: Normalized Discounted Cumulative Gain at position 3, measuring ranking quality of top results

Recall@20/100: Proportion of relevant documents retrieved in top 20/100 results

Generation Metrics: To evaluate the quality of generated responses and citation accuracy, we adopt the following metrics:

BLEU-1: Measures unigram overlap between generated and reference responses

ROUGE-L: Evaluates the longest common subsequence between generated and reference texts

Citation Labeling: Following ALCE[29], we evaluate the method's ability to accurately attribute information sources within generated responses. The generated response r_k consists of n statements $s_1, s_2, \ldots, s_n$. Each statement s_i cites a list of passages $C_i = \{c_{i,1}, c_{i,2}, \ldots\}$, where $c_{i,j} \in P_k$. We adopt Citation Recall and Citation Precision as defined in ALCE to evaluate the accuracy of citation labeling.

4.1.2Models and Retrieval Systems.

We experiment with Qwen2.5 language models of varying scales (0.5B, 1.5B, 3B, 7B parameters) for query reformulation and keyword extraction. For retrieval, we employ two complementary approaches: BM25, a sparse lexical retrieval method based on term frequency and inverse document frequency implemented using Faiss[30] and Pyserini[31]; And ANCE[32] (Approximate Nearest Neighbor Negative Contrastive Estimation), a dense neural retrieval model.

4.1.3Retrieval Strategies.

We evaluate four retrieval strategies to validate our attention-based keyword extraction:

1. Rewrite-only: Using only the LLM-reformulated query q'_n
2. Layer-Averaged Keywords: Keywords extracted using the optimal layer's averaged attention (Equation 4-5)
3. Expert Head Ensemble: Keywords from top-K expert attention heads (Equation 6-7)
4. Hybrid Strategy: Concatenation of reformulated query with extracted keywords (Equation 15-16)

4.1.4Context Compression Strategy.

To evaluate the effectiveness of our attention-based context compression for the generation phase (Equation 17), we implement:

Compressed Context Generation: Using only dialogue turns containing high-importance keywords as input context for answer generation, reducing context length while preserving retrieval-relevant information. As illustrated in Figure 2, dialogue turns highlighted in red boxes are retained, while turns with lower keyword importance scores are pruned. This selective compression maintains semantic coherence while significantly reducing context window requirements.

Figure 2. Attention-based Dialogue History Compression. Red-boxed dialogue turns containing high-importance keywords (e.g., "Cheryl", "arrest", "suspect") while other turns are pruned to reduce context length.

4.2 Overall Retrieval Performance

Table 1 presents comprehensive retrieval results for the Qwen-2.5 1.5B model across different retrieval methods and strategies. Several key observations emerge:

Table 1. Retrieval Performance of Qwen-2.5-1.5B on CORAL Dataset.

Retriever	Strategy	MRR	MAP	NDCG@3	Recall@20	Recall@100
BM25	rewrite	40.7	33.8	33.1	60.5	72.4
	Layer-Avg Keywords	42.9	36.2	35.1	65.1	76.8
	Expert Ensemble	44.6	37.9	36.7	67.1	79.3
	Hybrid	45.0	38.2	37.1	67.7	79.3
ANCE	rewrite	37.2	30.0	29.8	52.3	64.3
	Layer-Avg Keywords	35.2	28.0	27.8	50.0	62.3
	Expert Ensemble	36.8	29.4	29.1	52.4	65.0
	Hybrid	39.5	31.9	31.9	55.5	67.8

Key Findings:

1. Hybrid Strategy Superiority: The hybrid approach consistently outperforms all other strategies across all model scales and retrievers, validating our hypothesis that attention-extracted keywords provide complementary information to reformulated queries.

2. Expert Ensemble vs. Layer-Averaged: Expert head ensemble consistently outperforms single-layer averaged attention, demonstrating the benefit of identifying specialized attention heads. This suggests functional specialization exists within transformer attention mechanisms.

3. Dense vs. Sparse Retrieval: Interestingly, BM25 consistently outperforms ANCE across all settings in our experiments. This counterintuitive finding can be attributed to the nature of our attention-extracted keywords, which are exact lexical matches from the dialogue history. These keywords provide strong lexical signals that align perfectly with BM25's term-matching mechanism, while dense retrievers like ANCE may struggle to fully leverage these explicit

lexical cues. Notably, the relative improvements from our keyword extraction remain consistent (~10-15% relative gain) across both retrievers, indicating our method's robustness across different retrieval paradigms—though the synergy is particularly strong with sparse lexical methods.

4.3 Top-K Analysis

A critical hyperparameter in our method is K in Equation 8, which determines the number of keywords to extract from each output token's attention distribution. Figure 3 illustrates the impact of varying K values on retrieval performance using Qwen2.5-1.5B.

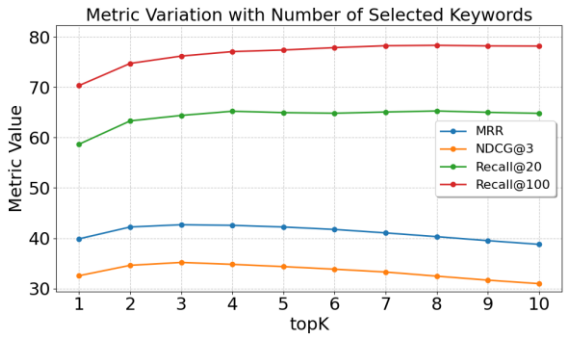

Figure 3. Retrieval Metrics vs. Number of Keywords per Token K.

Our experiments show that extracting top-3 keywords per output token achieves the best overall performance (NDCG@3: 35.2), providing an optimal balance between capturing sufficient context and avoiding noise. Beyond K=3, performance gradually declines as lower-ranked keywords introduce noise, while recall metrics plateau around K=5-7. This suggests that each output token's semantic focus can be adequately captured by a small set of highly-attended history tokens, aligning with cognitive science findings that humans typically focus on 3-5 key concepts when processing information.

4.4 Layer-wise Analysis

To understand how information relevant to keyword extraction flows through the transformer architecture, we evaluate each layer independently. Figure 4 shows retrieval performance using keywords extracted from different layers across four model scales.

Shallow Layer Instability and Middle Layer Optimality: Layers 0-5 exhibit high performance variance across models, suggesting early layers have not yet formed stable semantic representations suitable for retrieval. In contrast, all four models achieve peak performance in middle layers (approximately 60-70% depth): Qwen-0.5B at layer 20 (NDCG@3 = 33.8), Qwen-1.5B at layer 17 (NDCG@3 = 34.9), Qwen-3B at layer 22 (NDCG@3 = 35.7), and Qwen-7B at layer 15 (NDCG@3 = 36.4). This phenomenon occurs because early layers primarily encode low-level syntactic information, middle layers achieve optimal balance between semantic abstraction and fine-grained attention to specific tokens, while deep layers over-integrate information toward generation objectives, potentially losing discriminative attention patterns useful for retrieval.

Cross-Model Consistency and Limited Scale Advantage: Despite architectural differences, all models exhibit qualitatively similar layer-wise patterns with optimal layers occurring at 60-70% depth, suggesting this represents a

universal feature of transformer information processing. Notably, performance differences between 0.5B and 7B models are modest (2-3 absolute NDCG@3 points), indicating that attention-based keyword extraction is not a capability that strongly benefits from scale—even small models develop effective attention patterns for identifying relevant.

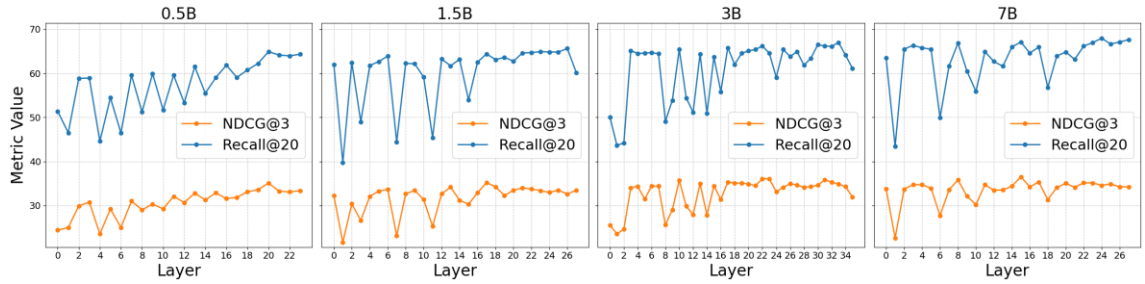

Figure 4. Layer-wise Keyword Extraction Performance.

4.5 Head-wise Analysis

Within each layer, different attention heads may specialize in different linguistic phenomena. Figure 5 compares head-level performance in the first layer (layer 0), optimal middle layer, and final layer across all four models.

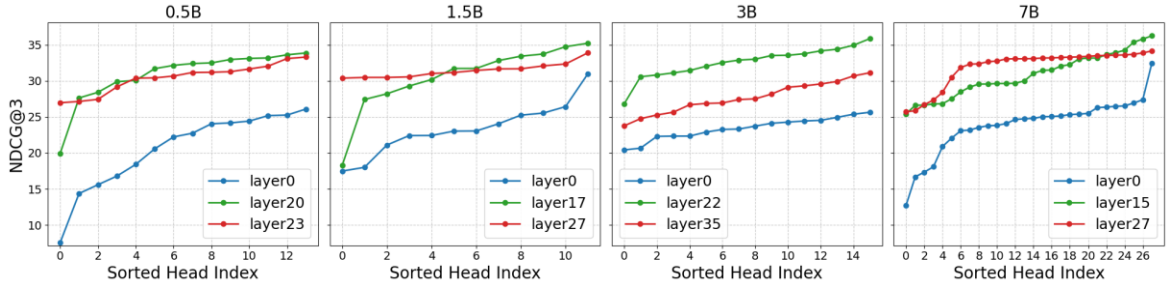

Figure 5. Head-wise Performance Distribution.

Substantial Head Heterogeneity and Functional Specialization: Heads within the same layer exhibit dramatically different retrieval effectiveness. For example, in Qwen-7B's layer 15, the best head achieves NDCG@3 = 35.2 while the worst only reaches 12.7—a variance of 22.5 points. This large performance gap demonstrates clear functional specialization, where certain heads have learned attention patterns specifically beneficial for identifying retrieval-relevant keywords. The existence of consistently high-performing heads across test examples motivates our expert ensemble strategy (Equation 7), effectively constructing a specialized "retrieval committee" within the broader transformer architecture.

Layer-Dependent Head Diversity: Layer 0 shows the widest head-to-head variation, with some heads performing near-randomly while others achieve moderate success, likely reflecting diverse low-level feature extraction strategies. Optimal middle layers display more uniform performance with several high-performing heads, suggesting convergence toward a common semantic representation space. Final layers show intermediate diversity—some heads maintain strong performance while others degrade, possibly due to specialization toward generation rather than comprehension tasks. Additionally, larger models (Qwen-7B) exhibit more pronounced head specialization than smaller ones (Qwen-0.5B),

suggesting that scale enables finer-grained division of labor among attention heads, though all scales contain identifiable expert heads.

4.6 Token-wise Analysis

4.6.1Attention Entropy and Token Selection.

Our hypothesis is that output tokens with diffused attention (high entropy) are more informative for keyword extraction than those with concentrated attention. Figure 6 validates this counterintuitive finding with Qwen2.5-1.5B.

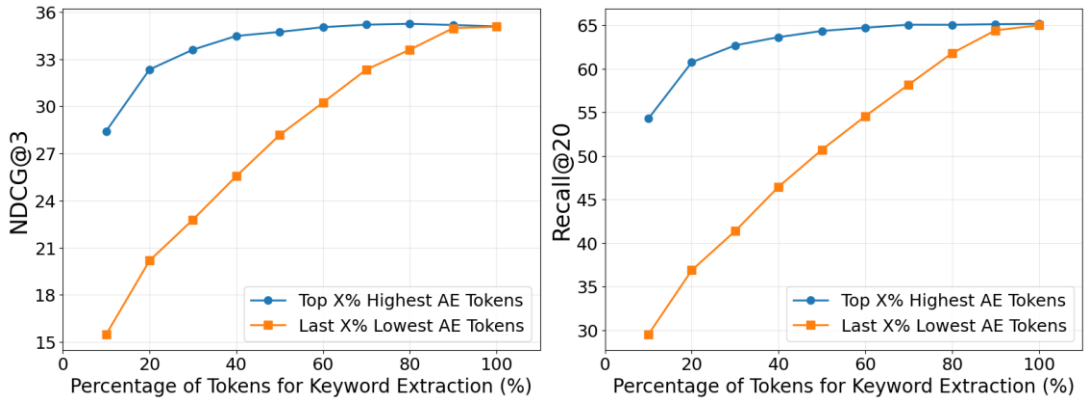

Figure 6. Retrieval Performance vs. Token Selection by Attention Entropy.

Selecting the top-X% highest-entropy tokens consistently achieves superior NDCG@3 and Recall@20 compared to lowest-entropy tokens, with optimal performance occurring at 50-70% for high-AE selection. This counterintuitive finding can be explained by the fundamental difference between information integration and information localization in token generation:

High-entropy tokens exhibit diffused attention across multiple historical tokens, representing information integration tokens that synthesize content from diverse context sources. These tokens—typically summarizing terms, relational concepts, or abstract semantic constructs—encode cross-turn dependencies and implicit information needs crucial for conversational retrieval. Low-entropy tokens, conversely, concentrate attention on a few specific tokens, acting as direct copy tokens that primarily depend on local information. These include entity names, numbers, or verbatim phrase repetitions—while locally accurate in reproducing explicit mentions, they miss the broader semantic associations and implicit contextual relationships necessary for comprehensive retrieval. In conversational search, effective keywords must capture not only explicitly mentioned entities but also implicit information requirements and cross-turn semantic connections. High-entropy tokens precisely capture these multi-turn semantic dependencies. Using the top 60% highest-AE tokens achieves peak NDCG@3 (35.8), balancing coverage (sufficient tokens to capture diverse semantics) and quality (filtering out overly specific copy tokens).

4.6.2Qualitative Analysis: Word Clouds.

To provide intuitive understanding of what distinguishes high vs. low attention entropy tokens, Figure 7 visualizes word clouds aggregated from the highest-AE and lowest-AE tokens across all reformulations in the test set.

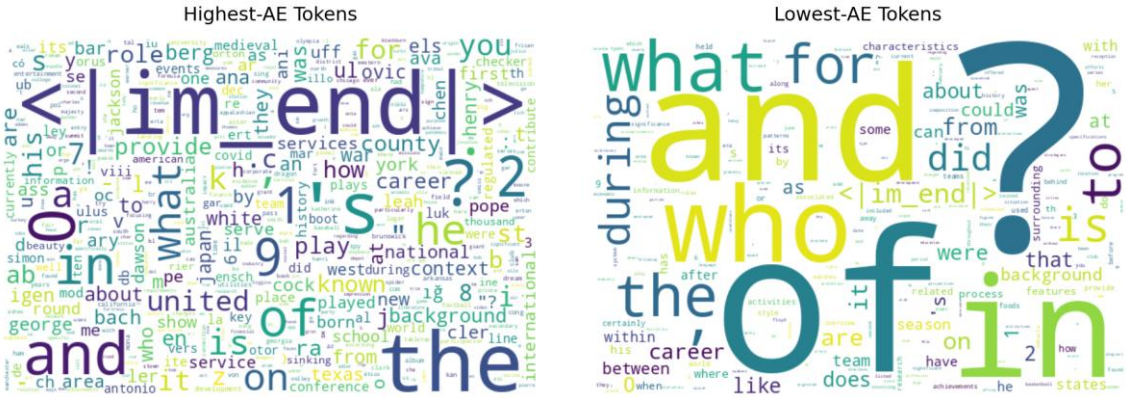

Figure 7. Word Clouds of High-AE vs. low-AE Tokens.

Left Panel (Highest-AE Tokens - Most Informative):

The word cloud reveals a rich distribution of content-heavy terms including proper nouns ("White", "Henry", "Texas", "Japan"), substantive verbs ("provide", "known", "born", "played"), concrete nouns ("services", "career", "school", "conference"), and complex concepts ("international", "professional", "background"). These terms carry high information density and semantic specificity, enabling precise retrieval when extracted as keywords.

Surprisingly, the end-of-sequence token '<|im_end|>' appears prominently among high-AE tokens. This reflects a profound insight into transformer decision-making from an information-theoretic perspective: generating EOS requires the model to verify multiple independent conditions across different regions of the dialogue history—Has the user's question been answered? Is any critical information missing? Is the logical chain complete? Is the semantics self-consistent? Is the format properly concluded? Each verification criterion necessitates examining distinct historical token regions, resulting in broadly diffused attention (high entropy). Thus, '<|im_end|>' serves as a global integration checkpoint that attends to the most salient keywords across the entire dialogue context to ensure comprehensive query reformulation. This finding suggests that structurally important tokens (EOS, delimiter tokens) naturally encode retrieval-relevant information hubs through their multi-faceted verification role.

Right Panel (Lowest-AE Tokens - Less Informative):

This word cloud is dominated by low-content function words including pronouns ("what", "who", "he", "his", "it"), prepositions ("of", "in", "to", "for"), copulas ("is", "was", "are", "were"), articles ("the", "a", "that"), and conjunctions ("and", "or"). These terms provide grammatical structure but minimal semantic content. Their low attention entropy reflects concentrated attention on specific syntactic positions without capturing diverse semantic information. Including such tokens as keywords introduces noise without improving retrieval precision, as they lack the cross-turn semantic integration necessary for effective retrieval.

4.7 Generation Quality Evaluation

To comprehensively evaluate our framework's effectiveness, we conduct generation quality experiments to assess whether our attention-based context compression strategy (Figure 2) preserves sufficient information for high-quality response generation.

We evaluate four context selection strategies using Qwen-7B as the generation model: Raw context uses the entire dialogue history $H = \{(q_1, r_1), (q_2, r_2), \ldots, (q_{n-1}, r_{n-1})\}$; Last Response heuristically selects all previous questions and

only the most recent response $f_{LR}(H_n) = \{q_i\}_{i=1}^{n-1}, r_{n-1}$; Rewrite uses only the LLM-reformulated query q'_n without historical context; and KBC (Keyword-based Compression) retains only dialogue turns containing high-importance keywords as determined by Equation 17. Table 2 presents generation quality metrics across different context selection strategies.

Table 2. Generation Quality on CORAL Dataset (Qwen-7B).

Strategy	BLEU-1	ROUGE-L	Citation Recall	Citation Precision
Raw Text	23.3	15.1	14.3	27.5
Last Response	24.1	15.8	14.5	29.1
Rewrite	21.7	13.9	12.5	30.6
KBC (Ours)	26.4	17.9	15.3	33.8

Our keyword-based compression strategy achieves the highest performance across all generation metrics, even outperforming the Raw Text baseline that uses the complete dialogue history. This demonstrates that our attention-guided selection effectively retains the most generation-relevant information while filtering out distracting or redundant content. The Rewrite baseline suffers from incomplete context and loses essential contextual nuances, while the Last Response heuristic achieves reasonable performance but misses critical information from earlier turns. Our method strikes an optimal balance by selecting dialogue turns based on attention-derived keyword importance scores that naturally preserve logical dependencies and discourse structure, while simultaneously achieving superior citation accuracy by preserving the source attribution chain necessary for accurate citation. In summary, our attention-based context compression strategy validates that attention patterns during query reformulation encode precisely the contextual information necessary for high-quality response generation.

5 CONCLUSION

In this paper, we proposed a novel attention-based framework for conversational retrieval-augmented generation that addresses the critical challenges of keyword omission and hallucination in query reformulation. Our key innovation lies in exploiting the intrinsic attention mechanisms of Large Language Models during query reformulation to identify and extract salient keywords from dialogue history, thereby bridging the gap between reformulation quality and retrieval effectiveness. By leveraging already-computed attention weights during LLM inference, our method introduces zero additional computational cost while achieving substantial performance improvements over reformulation-only baselines. Through systematic layer-wise, head-wise, and token-wise analyses, we provide valuable insights into how LLMs process conversational context, revealing that middle layers achieve optimal keyword extraction performance, substantial functional specialization exists among attention heads, and counterintuitively, output tokens with diffused attention are more informative for retrieval than those with concentrated attention as they capture multi-turn semantic dependencies rather than local lexical copying. Our attention-based context compression strategy not only reduces context length but actually improves generation quality, outperforming complete dialogue history across both retrieval and generation metrics including citation accuracy. This work demonstrates that the internal attention mechanisms of LLMs contain rich and actionable signals for improving conversational retrieval systems, and we believe our attention-driven framework represents a promising direction for developing more effective, efficient, and interpretable conversational AI systems by systematically analyzing and exploiting these attention patterns without architectural modifications or additional training overhead.

Imitations and future work. Despite the promising results of our attention-based framework for conversational retrieval, several limitations should be acknowledged. Our approach relies heavily on the quality and characteristics of the underlying LLM's attention mechanisms, which may vary significantly across different model architectures and sizes. The generalizability of our findings across diverse languages, specialized domains, and conversational styles remains to be thoroughly validated, as the optimal layer selection and attention head configurations identified in our study may not transfer directly to all scenarios. Additionally, our current framework applies uniform keyword extraction strategies across all conversational turns, without dynamically adapting to varying dialogue complexity, topic shifts, or information density. The relationship between attention patterns and retrieval effectiveness is not always straightforward, and our method may struggle with highly ambiguous queries or those requiring complex reasoning across multiple distant turns where subtle long-range contextual cues are essential. Several promising directions for future research emerge from this work. First, developing adaptive mechanisms to automatically select optimal layers and attention heads based on query characteristics and dialogue complexity could further improve performance. Second, extending our framework to multi-modal conversational scenarios and investigating how cross-modal attention patterns inform retrieval represents an exciting frontier. Third, exploring how attention patterns can directly guide retrieval ranking algorithms and establishing a more rigorous theoretical foundation connecting attention mechanisms with retrieval effectiveness would provide principled design guidelines. Finally, examining robustness against adversarial inputs and integrating our approach with emerging LLM architectures will ensure the framework remains effective as the field evolves. By addressing these limitations and pursuing these research directions, we envision more effective, efficient, and interpretable conversational retrieval systems that better serve user needs.

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